Program: **Diploma in Food Processing Technology**	
Course Code: **6428**	Course Title **Dairy Processing Technology Lab**
Semester: **6**	Credits: **2.5**
Course Category: **Program core**	
Periods per week: **4 (L:0 T:1 P:3)**	Periods per semester: **60**

Course Objectives:

- Determine the microbiological quality of raw milk
- Prepare certain dairy products
- Determine key parameters of various dairy products

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Awareness about milk product		Dairy processing technology	6

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Apply various tests to determine the quality of raw milk, including resazurin test, lactometer reading, pH and acidity, SNF content, and specific gravity.	20	Applying
CO2	Apply techniques for determining the microbiological quality (TPC/SPC) of pasteurized milk, sterilized milk, and ice cream, as well as assess the adequacy of pasteurization.	20	Applying
CO3	Apply techniques to prepare flavored milk, yogurt, and ice cream in a laboratory setting.	10	Applying

CO4	Determine key parameters in dairy products, such as overrun in ice cream, and apply knowledge gained from a visit to a dairy or ice cream factory.	10	Applying

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped,2-Moderately mapped,1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Blooms Taxonomy Level
CO1	**Apply various tests to determine the quality of raw milk, including resazurin test, lactometer reading, pH and acidity, SNF content, and specific gravity.**		
M1.01	Determination of quality of raw milk by resauzurin test	4	Applying
M1.02	Determination of quality of raw milk by lactometer reading	4	Applying
M1.03	Determination of quality of raw milkusing pH and acidity	4	Applying
M1.04	Determination of quality of raw milk based on SNF content	4	Applying
M1.05	Determination of quality of raw milk by specific gravity	4	Applying
CO2	**Apply techniques for determining the microbiological quality (TPC/SPC) of pasteurized milk, sterilized milk, and ice cream, as well as assess the adequacy of pasteurization.**		
M2.01	Determination of microbiological quality (TPC/SPC) of pasteurized milk sample	4	Applying

M2.02	Determination of microbiological quality (TPC/SPC) of sterilized milk sample	4	Applying
M2.03	Determination of adequacy of pasteurization.	4	Applying
M2.04	Determination of microbiological quality (TPC/SPC) of ice cream	6	Applying
	Lab Test –I	2	
CO3	**Apply techniques to prepare flavored milk, yogurt, and ice cream in a laboratory setting.**		
M3.01	Apply the techniques and procedures to prepare various dairy products, specifically flavoured milk, by following standard guidelines and utilizing appropriate tools and ingredients	3	Applying
M3.02	Demonstrate the ability to prepare yogurt by applying appropriate techniques and procedures for the production of dairy products.	3	Applying
M3.03	Apply the principles and techniques of dairy product preparation to produce ice cream, by selecting appropriate ingredients, and managing the freezing process effectively.	4	Applying
CO4	**Determine key parameters in dairy products, such as overrun in ice cream, and apply knowledge gained from a visit to a dairy or ice cream factory.**		
M4.01	Apply the principles of overrun determination to evaluate key parameters in dairy products, specifically ice cream.	4	Applying
M4.02	Apply their knowledge of dairy/ice cream production processes by visiting a factory and demonstrating the practical use of production techniques, quality control methods, and machinery operation in the context of the dairy/ice cream industry.	4	Applying
	Lab Test– II	2	

Text/Reference

T/R	Book Title/Author
R1	Tufail Ahmed, "Dairy Plant Engineering and Management", CBS Publishers and Distributors, New Delhi, 2001, ISBN-13: 978-8122501186

R2	Pieter Walstra, Jan T M Wouters, Tom J Geurts, “Dairy Science and Technology”, Taylor Francis Group, CRC press. 2006. ISBN-13: 978-0824727635
R3	De Sukumar, “Outlines Of Dairy Technology”, Oxford University Press.2015, ISBN-13: 978- 0195611946
	Spreer, Edgar “Milk and Dairy Product Technology”. Marcel Dekker, 2005. ISBN-13: 978-0824700942

Online Resources

Sl.No	Website Link
1.	https://dairynutrition.ca/en/milk-quality/homogenization/why-milk-homogenized-and-what-are-its-effects
2.	https://khatabook.com/blog/how-to-manufacture-butter-step-by-step-process/
3.	https://thecheeseweb.com/7-types-of-cheese
4.	https://www.sciencedirect.com/science/article/pii/S0022030279831975